

SAGE-MM: coordinating heap, interop, and page reclamation under one embedded memory budget

Problem



one swapless process budget

Tensions: heap ↔ footprint,
allocation ↔ interop,
residency ↔ refault. Tuned in
isolation = globally fragile.

SAGE-MM coordination

Heap build (**G**)
build-time

Interop (**I**)
source-time

static: reduce the load

Online controller (C)
telemetry ⇒ pressure
reclamation interval T
+ compaction gate
hysteresis · cooldown · fail-closed

only this adapts online

Measured: ARM32/ARM64, $n=30$

peak PSS **-24%**

GC tail **-37%**

refault **recovered**

ctrl CPU $\approx 1\%$

Robust across regimes

- **Favorable:** clear win
- **Neutral:** within noise (fail-safe)
- **Adverse:** bounded, guarded regression

*Every number: per-run, 95% boot-
strap CI, fail-closed pipeline.*